Write Blocker Tools

What Is a Write Blocker?

A **write blocker** is a critical forensic tool that allows an investigator to access a digital storage device — USB drive, hard disk, or SSD — in **read-only mode**, preventing any write operations from reaching the original media.

This ensures the evidence remains unaltered during acquisition, preserving both **data integrity** and **legal admissibility**.

Core Principle: No bit on the original evidence device should ever be modified during examination. Even a single write — such as an auto-mount — can change metadata, timestamps, or system state, undermining forensic integrity.

Why Write Blockers Are Non-Negotiable

Integrity

Ensures the original device remains completely unmodified throughout examination.

Repeatability

Enables re-acquisition that yields bit-for-bit identical results every time.

Legal Admissibility

Courts require demonstrable evidence integrity; write blockers provide it.

Chain of Custody

Read-only access maintains a defensible, auditable provenance record.

> Real case reports frequently cite **improper write blocking** as grounds for evidence exclusion in court proceedings.

Hardware Write Blockers

Hardware blockers physically sit between the evidence media and the forensic workstation, intercepting and filtering commands to permit **only read operations**. They are vendor-agnostic, file-system independent, and the most forensically defensible option.

1

Tableau Forensic Imager Series

Models T8i, T35u, T356789uTX. Supports USB, SATA, and IDE. Industry standard in law enforcement labs globally with on-unit imaging and hashing.

2

WiebeTech Write Blockers

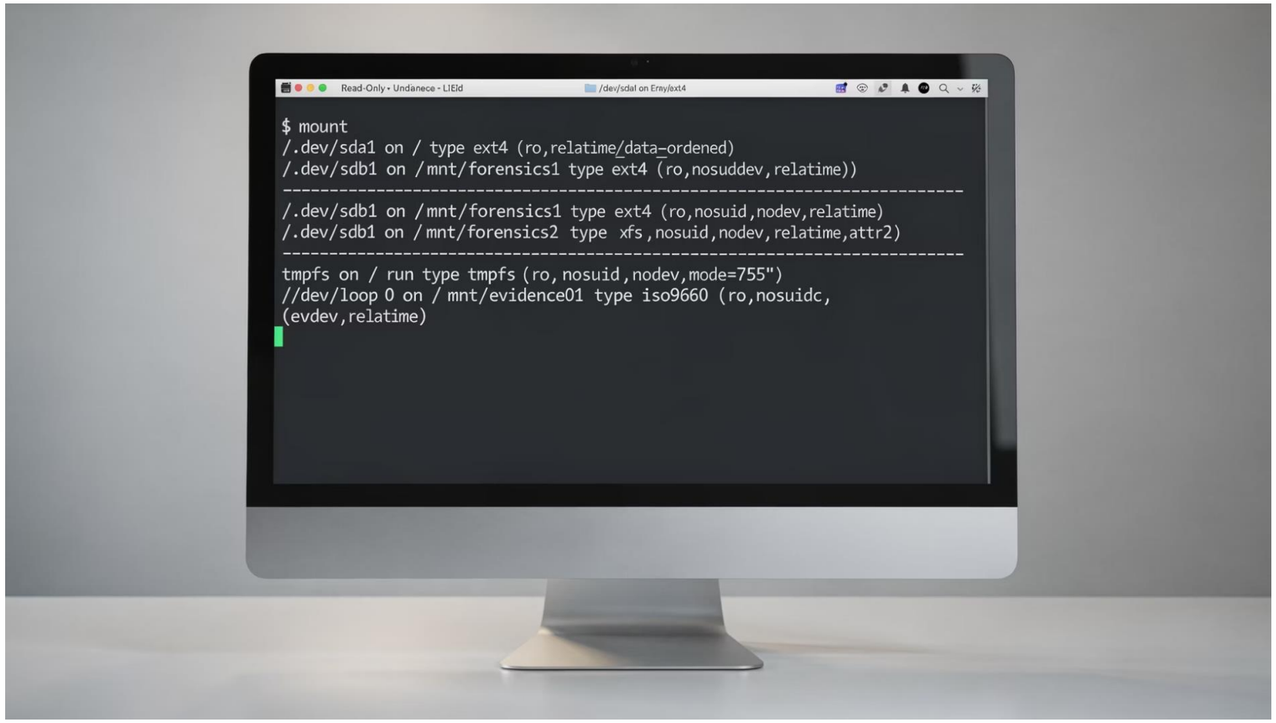
Popular in corporate and private labs. Often modular or rack-mountable for flexible deployment across investigation teams.

3

Logicube Forensic Duplicator

Standalone imager with integrated write blocking. Particularly useful in mass imaging scenarios requiring high throughput.

Software Write Blockers



Software blockers are installed at OS level and intercept file system calls to prevent writes. They are **low cost or free**, but carry notable limitations.

OSFMount (Windows)

Mounts images read-only, protecting against accidental writes.

Linux Mount Options

`mount -o ro,noload` — requires care to disable automount.

Suite-Integrated Drivers

Axiom and FTK include built-in software write-blocking modules.

Software blockers can be bypassed by kernel-level processes and are OS/file-system specific. **Not ideal for court-bound evidence** unless paired with hardware controls.

Acquisition Workflow With Write Blockers

This seven-step workflow is cited in forensic examiners' deposition scripts and forms the backbone of defensible evidence collection.

Document Device

Record serial, model, and photos

Connect with Blocker

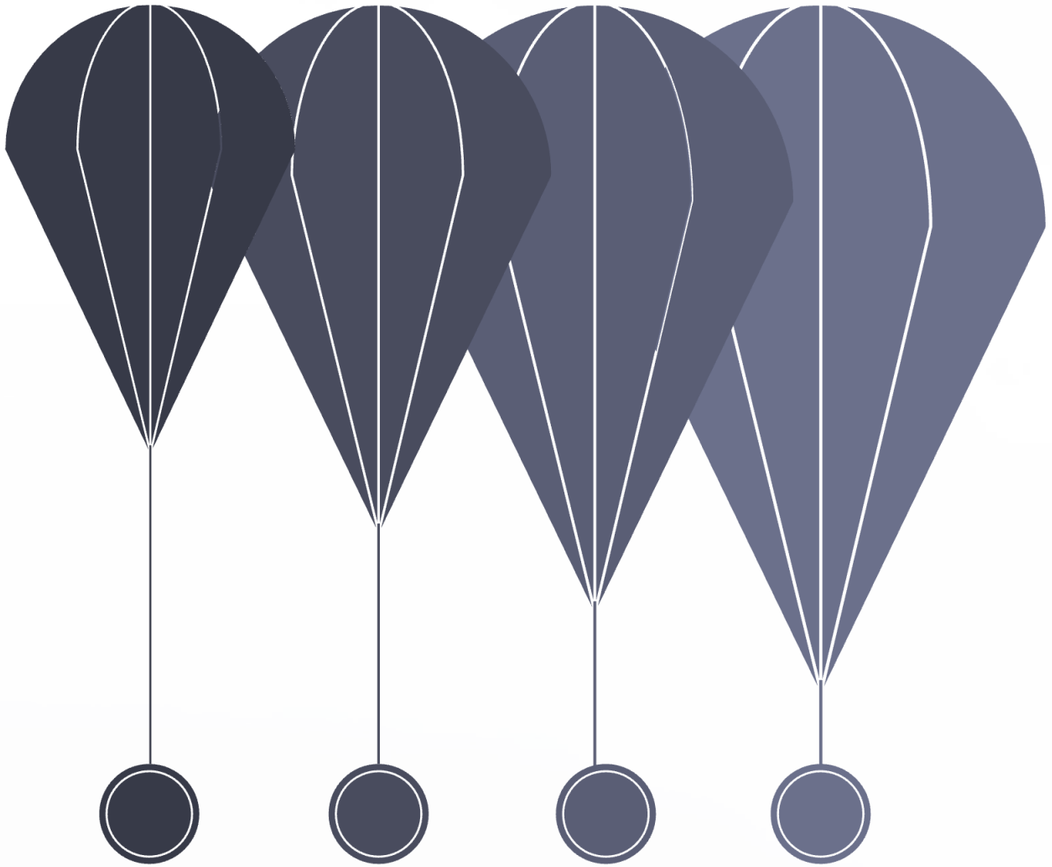
Use hardware write blocker

Verify Read-Only

Confirm device is non-writable

Forensic Imaging

Acquire image (FTK Imager or dd)



Each step must be meticulously documented. Skipping or reordering steps — particularly hashing — creates vulnerabilities that opposing counsel can exploit.

Examples of Write Blocker Tools

Tool	Type	Interfaces	Key Feature	Typical Use Case
Tableau T8i	Hardware	SATA / USB	Industry standard	Law enforcement labs
Tableau T35u	Hardware	SATA / IDE / USB	Imaging + hashing	Corporate forensic teams
Logicube Duplicator	Hardware	SATA / IDE / USB	High throughput	Mass device acquisition
WiebeTech USB	Hardware	USB	Simple protection	Small labs / consultants
OSFMount	Software	Virtual mount	Read-only images	Preliminary analysis
Linux mount -o ro	Software	OS-level	Free / flexible	CLI environments

Real-World Pitfalls & How to Avoid Them

Even experienced examiners make critical errors. The following four pitfalls are drawn from documented cases and peer-reviewed literature.

Pitfall 1 — Assuming "Read-Only" Is Enough

The Mistake

Plugging suspect USB media directly into a forensic workstation without a hardware write blocker.

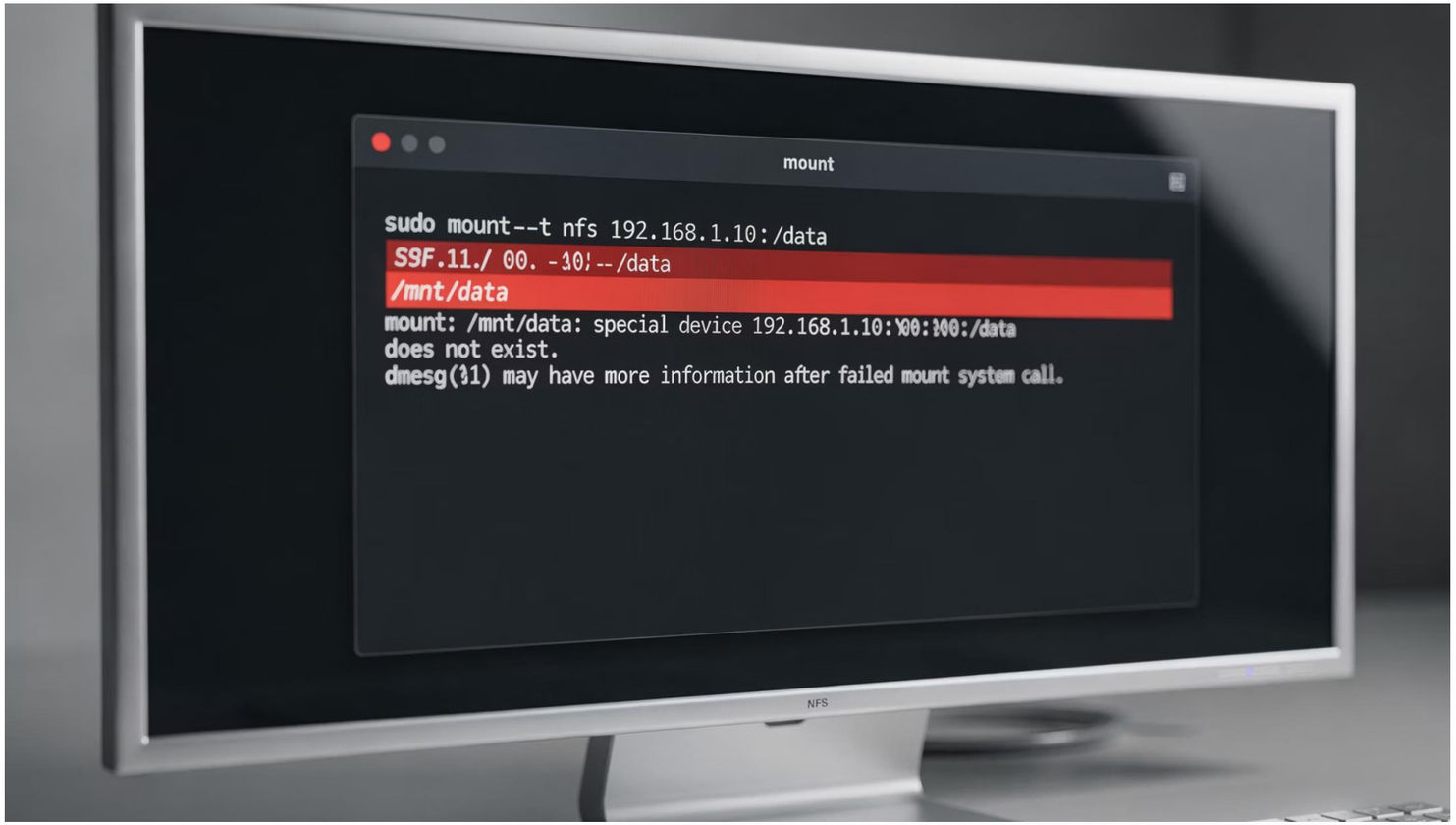
The Consequence

OS updates, metadata writes, thumbnail caches, and access timestamps change — compromising evidence.

The Solution

Always insert suspect media through a hardware write blocker. Never connect directly.

Pitfall 2 : Software Blocker Misconfiguration



The Problem

Linux `mount -o ro` does **not** prevent all write attempts unless `noload` and proper options are explicitly set

Noload prevents the system from loading/replaying the filesystem journal during the mount process

The Risk

Hidden background writes could occur silently, tainting evidence without the examiner's knowledge.

The Fix

Prefer **hardware write blockers**. If software-only, verify mounts are truly non-writable via controlled write-attempt tests before proceeding.

Pitfall 3 : Incorrect Imaging Settings

1

The Error

Skipping hash generation **during** acquisition — imaging first, hashing later.

2

The Risk

Image and original may not match. Undetected errors create a viable legal challenge.

3

Best Practice

Generate MD5 + SHA-256 hashes **in real time** during acquisition using FTK Imager or dc3dd.

Pitfall 4 : Failure to Document Blocker Details

The Oversight

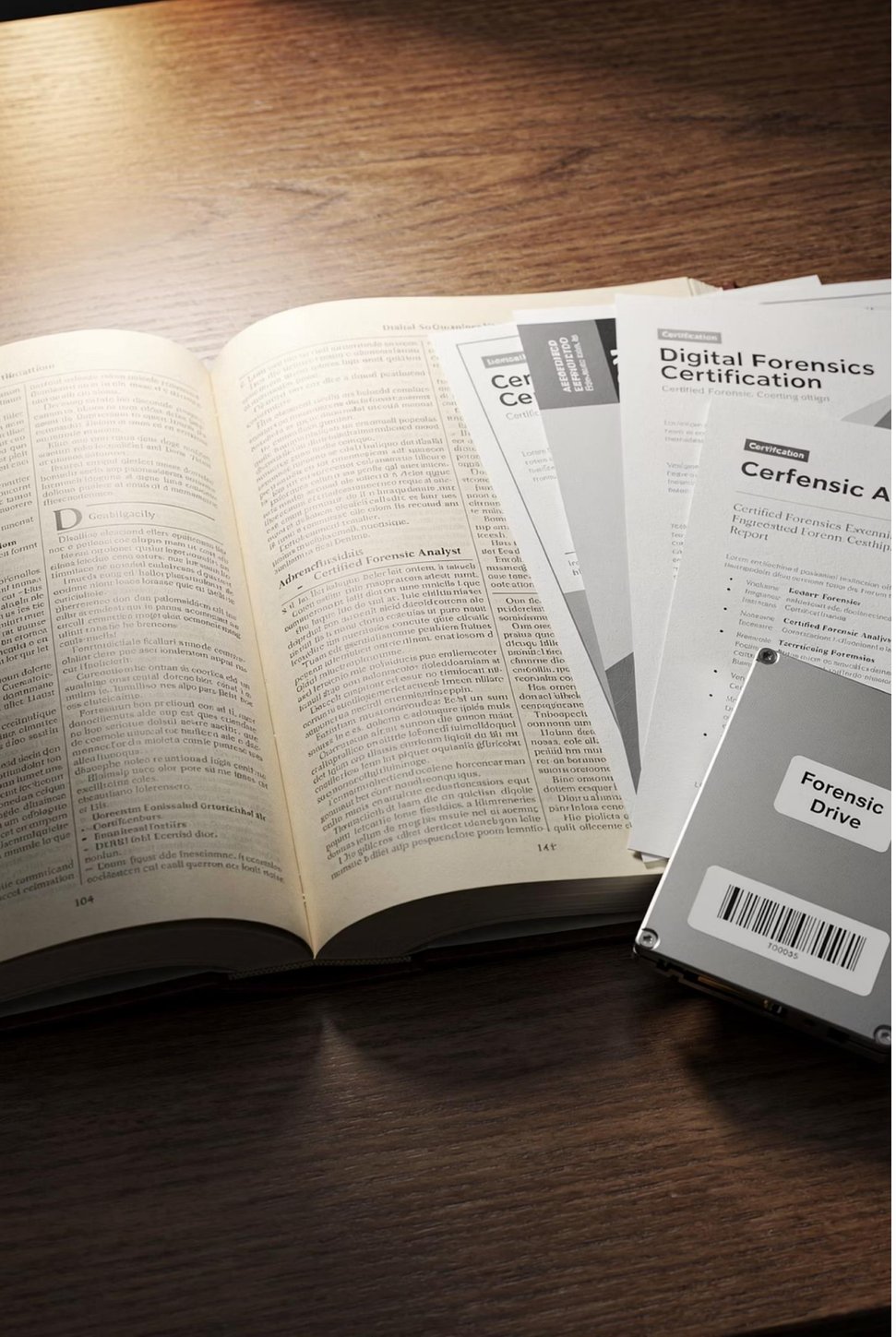
Chain-of-custody records that omit write blocker model, serial number, and connection details are fundamentally weak.

Defence Counsel Angle

Without device-specific documentation, opposing counsel can challenge the credibility of the entire acquisition — questioning whether write-blocking was actually in place.

The Standard

- Photograph the write blocker in situ
- Record make, model, serial number, and firmware version
- Document the connection diagram and cable configuration



Case Reference Evidence

> "Digital evidence was acquired using a Tableau T8i write blocker. Hashing algorithms MD5 and SHA-256 were computed during imaging."
>
> — Standard phrasing in published law enforcement forensic examination reports

Law Enforcement Disclosures

Published forensic exam reports routinely cite the specific write blocker model and hashing methodology used during acquisition.

Certification Standards

SANS, GCFA, and EnCE certification courses all emphasise **hardware blocking + hashing** as core, non-negotiable standards.

Academic Literature

Forensic textbooks consistently identify hardware write blockers as the gold standard for preserving evidentiary integrity.

1

Write Blockers Are Essential

They protect both evidence integrity and legal admissibility. There is no acceptable substitute in professional practice.

2

Hardware Over Software

Hardware write blockers are preferred for court-ready evidence. Software blockers may support analysis but are weaker defensively.

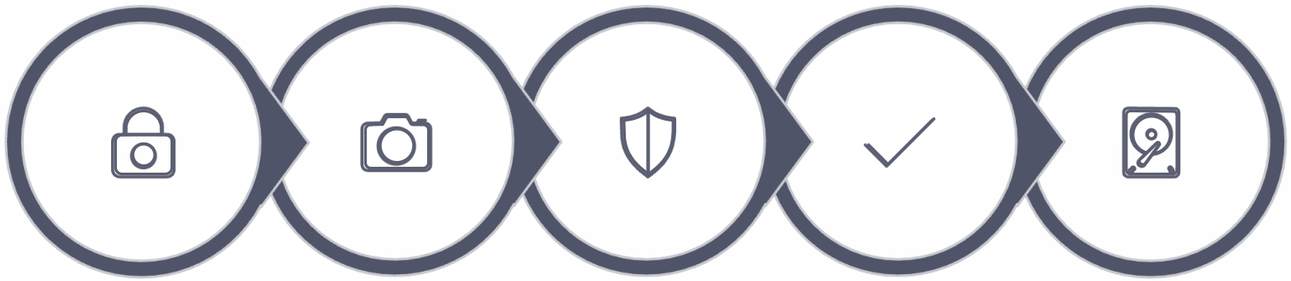
3

Document Everything

Clear documentation and real-time hashing during acquisition are equally critical to the forensic process.

When executed correctly, this process forms a **forensically sound chain of custody** that will withstand rigorous legal scrutiny.

Forensic Acquisition Process — Visual Reference



Secure Evidence | Photograph Device | Attach Write Blocker | Verify Read-Only | Create Image

This visual reference is designed for inclusion in **Standard Operating Procedures (SOPs)** and forensic training materials. Print or embed it alongside your lab's chain-of-custody forms for a complete, audit-ready workflow.